Status	Finished
Started	Saturday, 17 October 2026, 6:02 PM
Completed	Saturday, 17 October 2026, 6:05 PM
Duration	2 mins 25 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data in a single pass
- c. Sorting data sequentially
- d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection
b. Extraction
c. Insertion
d. Distribution

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Searching
c. Selection
d. All of the above

The correct answer is: All of the above

Question **5**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Bubbling up the largest element to its correct position with each pass
c. Sorting data in a non-sequential manner
d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are compared element by element
c. They are discarded
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Provides a basis for other algorithms to work efficiently
c. Makes code execution slower
d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Bubble Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It does not require recursion
- c. It is less efficient than bubble sort
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **12**

Complete

Mark 0.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Divide and conquer
c. Backtracking
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Linear Search
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Arranging data in a particular format
c. Deleting data from a list
d. Searching for data in a list

The correct answer is: Arranging data in a particular format